Clebsch: Rigidity from Sparse Shadows — **V**

The Golden Companion Correspondence

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August 2026

From deep holes, the cubic takes shape, finds its bearings,
stands fixed while its shadows move; from minimum words, the plane returns,
and ***the scattered shadows gather home.***

Abstract

Paper II of this series produces a signed cubic in a ten-dimensional matching quotient. We identify its canonical five-dimensional residue and prove that it is not the conference cubic appearing in Papers I and III: the two are distinguished companions in the two-dimensional pencil invariant under the irreducible five-dimensional action of A_5. The conference member has six nodes. The Paper-II residue is a chordal Hankel determinant, singular along a rational normal quartic. Its twelve rational singular points form A_5/C_5; pairing points with the same stabilizer gives A_5/D_5 and identifies its six fibres equivariantly with the original matched axes. Thus the two companion cubics realize one six-axis carrier through different singular shadows. We determine the normalized outer-normalizer action exactly on the Paper-II pencil and prove that, after one chordal companion is retained, its difference operator gives an exact oriented round trip between sheet and conference orientations. The result rules out a tempting literal identification of the two invariant cubic lines while recovering the marked equivalence that replaces it.

Keywords. icosahedral invariant; chordal cubic; rational normal quartic; conference matrix; perfect matching; reconstruction.

2020 Mathematics Subject Classification. 14J30, 20C20, 05C70, 51E20.

1 Introduction and statement

An invariant can retain the correct symmetry and still be the wrong invariant. The five-dimensional irreducible representation of A_5 has a two-dimensional space of invariant cubics, not a unique invariant line. Consequently, agreement of the carrier, group action, and orientation character does not identify the signed cubic extracted from a matching quotient with the triangle cubic of a conference matrix. Papers I and III construct the latter with explicit markings [7, 8].

The exceptional six-point actions, matching dictionaries, and Joubert–Segre cubics are classical; we follow Howard–Millson–Snowden–Vakil [4, §§1.1–1.6 and 2.1–2.4]. Conference switching is likewise standard [3, §§1–3]. Our contribution is to locate and normalize the specific Paper-II tensor in this pencil over $\mathbf{F}_{11}$, then identify its marked axes.

Fix the H_3 matching

$$M = \{01, 25, 37, 49, 68, 10\,11\}$$

on the twelve points of $\mathbf{P}^1(\mathbf{F}_{11})$. Its stabilizer in $\mathrm{PSL}_2(11)$ is a group $G \cong A_5$, acting transitively on the six pairs of M. Write X_M for that six-set and

$$A_M = \{(u_x)_{x\in X_M} \in \mathbf{F}_{11}^{X_M} : \textstyle\sum_x u_x = 0\}$$

for its augmentation module. Paper II constructs a ten-dimensional quotient W, an intrinsic pair of matching sheets, and their first nonzero signed moment $\mu_3 \in \mathrm{Sym}^3(W)$ [6]. Since the action on X_M is transitive, a pair stabilizer has order ten; the order-ten subgroups of A_5 are the self-normalizing groups D_5. Thus $X_M \cong A_5/D_5$.

Let W_5 be the five-isotypic summand of $W|_G$. The invariant self-duality of W_5 turns the projection of μ_3 into a projective cubic line $[H_M] \subset \mathbf{P}(\mathrm{Sym}^3(W_5^*))$.

Definition 1.1 (Retained chordal residue)**.** The retained Paper-II input is the triple

$$\mathcal{R}_M = (G \curvearrowright W_5, [H_M]).$$

It remembers the marked G-action and the projective cubic line. It does not remember X_M, an axis basis, a conference signing, or a generator of $[H_M]$. Isomorphisms are G-equivariant projectivities carrying the cubic line to the cubic line.

Theorem 1.2 (Chordal placement and axis compatibility)**.** *The marked singular geometry of $\mathcal{R}_M$ supplies a canonical six-element G-set $\widehat{X}_M$. There is a unique G-equivariant identification $\widehat{X}_M \cong X_M$.*

More explicitly, after precomposing the G-action with the nontrivial outer automorphism of A_5, the retained cubic has equation

$$H = z_0z_2z_4 + 2z_1z_2z_3 - z_0z_3^2 - z_1^2z_4 - z_2^3 = \det\begin{pmatrix} z_0 & z_1 & z_2 \\ z_1 & z_2 & z_3 \\ z_2 & z_3 & z_4 \end{pmatrix}.$$

Its singular locus is the rational normal quartic

$$R: \ [s:t] \longmapsto [s^4 : s^3t : s^2t^2 : st^3 : t^4].$$

The twelve points $R(\mathbf{F}_{11})$ form the homogeneous G-set A_5/C_5. Pairing two points precisely when their exact stabilizers in G agree gives $\widehat{X}_M \cong A_5/D_5$.

The recovered six-set has a unique G-invariant order-six conference switching class up to opposite. Its triangle cubic defines a projective line $[C_M]$ in the same invariant pencil. The lines $[H_M]$ and $[C_M]$ are distinct. Thus $\mathcal{R}_M$ recovers both the original six-axis carrier and its unoriented conference companion.

Abstractly, the six-set A_5/D_5 is already determined by the retained group G, through its six Sylow-5 normalizers. The theorem's content is that the stabilizer fibres cut out by the Paper-II chordal singular geometry give exactly that six-set and agree with the original matching pairs.

The invariant pencil was previously exhibited in the needed characteristic-zero model: Pinardin and Zhang write its two generators and identify its two chordal members and singular rational normal quartics [5, §6.2, equations (6.3)–(6.5)]. The content of Theorem 1.2 is the exact placement of the Paper-II signed tensor in the reduction at eleven and the marked recovery of its original six axes. The A_5 marking is essential: an unmarked chordal cubic has automorphism group PGL_2[2, §5].

Proposition 1.3 (Outer action on the companion pencil)**.** *Order the six pairs as displayed above, use the induced lexicographic cubic basis, and normalize generators C_M, H_M so that their first nonzero coefficients are one. In this ordered pencil basis, an odd element of the order-*120 *normalizer of G on the six axes acts by*

$$[a : b] \longmapsto [-a + 8b : b].$$

It reverses the chosen conference generator, fixes the ten-point member $[1 : 3]$*, and exchanges the two chordal members* $[0 : 1]$ *and* $[1 : 7]$*.*

The selected chordal line is essential in the reverse direction: the conference package alone does not distinguish the two chordal companions. Once that line is retained, however, the same outer action supplies the orientation comparison rather than merely obstructing it. Pinardin and Zhang already exhibit the nonstandard S_5-extension and its involution exchanging the two rational normal quartics in characteristic zero [5, §8.2]. The assertion below concerns the exact normalized action on the Paper-II reduction and its compatibility with the two source orientation torsors.

Theorem 1.4 (Marked oriented companion round trip)**.** *Let*

$$\mathcal{I} = \mathrm{Sym}^3(A_M^*)^G$$

and let q denote the linear action on $\mathcal{I}$ induced by the nontrivial coset of $N_{S_6}(G)/G$. This action is independent of the coset representative. Put $\Delta = q - 1$.

The conference line is the (-1)-eigenspace $\mathcal{I}^-$ of q. For either chordal line L, the restriction

$$\Delta|_L : L \longrightarrow \mathcal{I}^-$$

is an isomorphism. In the normalization of Proposition 1.3, the forward and reverse maps are

$$h \longmapsto c = 8^{-1}\Delta h, \qquad c \longmapsto h = (\Delta|_L)^{-1}(8c).$$

They are inverse. Moreover $-h \mapsto -c$, so the map identifies the Paper-II sheet-orientation torsor with the conference-orientation torsor.

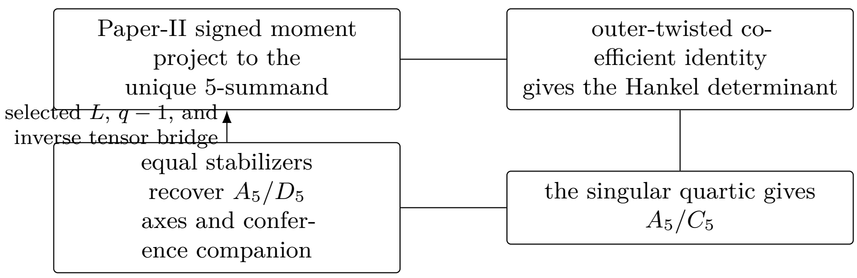
Consequently, the marked packages

$$(G \curvearrowright A_M, L, h) \quad \longleftrightarrow \quad (G \curvearrowright A_M, L, c)$$

are exactly recoverable in both directions. The line L is returned as part of the marking; no claim is made that an unmarked conference cubic chooses between the two chordal companions.

Thus the missing datum is not an arbitrary bijection between two abstract two-element sets. It is precisely the choice of chordal companion in the marked invariant pencil. Paper II supplies that choice through its projected tensor; a transport starting from Papers I or III must declare it as part of its bridge marking.

Proof and reading map.



Section 2 proves the projection and coefficient identity. Section 3 gives the human singular-locus and stabilizer proof. Section 4 proves the marked oriented companion theorem while isolating its exact unmarked boundary. Section 5 types the three source-specific returns. Paper IV is the independent branch of the series: its minimum-word reconstruction is not used here.

2 The five-dimensional residue

The restriction of Paper II's quotient to G has character

class	1	2	3	$5A$	$5B$
class size	1	15	20	12	12
χ_W	10	2	1	0	0

and hence

$$W|_G \cong \mathbf{1} \oplus V_4 \oplus V_5.$$

The quotient and signed moment are imported from Paper II's stable results *The* $3, 6, 10$ *quotient ranks* and *Balanced sheets and cubic-first orientation* [6]; the displayed restriction is the character calculation used here. The field $\mathbf{F}_{11}$ contains the fifth roots of unity and $\sqrt{5} = 4$, and $11 \nmid 60$, so it is a splitting field for this calculation and the summands are absolutely irreducible. Because $11 \nmid 60$, the central character idempotent

$$e_5 = \frac{5}{60} \sum_{g \in G} \chi_5(g) g$$

is defined over $\mathbf{F}_{11}$, is idempotent, and has rank five. The degree-six permutation character is $\mathbf{1} + \chi_5$, so $A_M \cong V_5$. Schur's lemma gives

$$\dim_{\mathbf{F}_{11}} \operatorname{Hom}_G(A_M, e_5 W) = 1.$$

Proof mode for the transport. The representation-theoretic reduction is the human argument above. Both A_M and $e_5 W$ are copies of the absolutely irreducible module V_5, so their intertwiner is unique up to scalar. There are two comparisons with the binary-quartic model: the direct identification and its outer twist. The singular-curve test selects the twist; the matrices below then fix its scalar and compare the cubic coefficients. These finite calculations are computer-assisted identities over $\mathbf{F}_{11}$. They are part of the proof, not merely discovery data. The versioned standard-library checker and its exact JSON output are named in Section 6; it rebuilds both carrier actions from Paper II's frozen inputs, solves the generator equations, and compares all thirty-five cubic coefficients.

For auditability, choose the axis basis $e_i - e_6$ of A_M and Paper II's reduced-pivot basis of W_5. One generator of the intertwiner line $A_M \to W_5$ is

$$T = \begin{pmatrix} 2 & 9 & 7 & 6 & 4 \\ 0 & 9 & 10 & 10 & 6 \\ 5 & 1 & 9 & 3 & 0 \\ 0 & 3 & 10 & 1 & 6 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Substitution in generators of G verifies $\rho_{W_5}(g)T = T\rho_{A_M}(g)$; its determinant is nonzero. Put $B = I + J$, where J is the 5-by-5 all-ones matrix; this is the invariant self-duality in the axis basis. Let Pol denote the standard conversion from a symmetric tensor to its homogeneous polynomial, including the mixed-monomial multiplicities. If $\nu \in \mathrm{Sym}^3(W_5)$ is the chosen-sheet projection of μ_3, its axis-cubic coefficient vector is

$$\widetilde{h}_M = \mathrm{Pol}\Big(\mathrm{Sym}^3(BT^{-1})\nu\Big) .$$

The direct Sym^4-identification has the right character but does not carry the Veronese curve to the singular curve. After the nontrivial class of $\mathrm{Out}(A_5)$, the required projectivity is

$$U = \begin{pmatrix} 0 & 0 & 3 & 5 & 3 \\ 4 & 1 & 1 & 10 & 4 \\ 3 & 6 & 3 & 0 & 0 \\ 5 & 1 & 6 & 10 & 5 \\ 9 & 10 & 10 & 7 & 1 \end{pmatrix} .$$

Both the direct and twisted intertwiner spaces are one-dimensional, but only the twisted one has the geometric property just stated. The first nonzero coefficient of the polynomialized vector $\widetilde{h}_M$ is 3. Put

$$h_M = 3^{-1}\widetilde{h}_M = 4\widetilde{h}_M .$$

The factor 3^{-1} is held fixed when the sheet is reversed, so the other sheet gives $-h_M$, not a second projective normalization. With the pullback convention used by the checker, $U(h_M)$ means $h_M(Uz)$. All its coefficients vanish except

$$[z_0z_2z_4, z_1z_2z_3, z_0z_3^2, z_1^2z_4, z_2^3] = [8, 5, 3, 3, 3].$$

This is exactly $8H$. Thus the normalized generator used in Theorem 1.4 is h_M, while $3h_M$ is its polynomialized axis image before Pol^{-1}, inverse self-duality, and the inverse bridge recover the projected Paper-II tensor. The conference coefficient line is different; the two normalized coefficient vectors have rank two. This proves the chordal-placement assertion of Theorem 1.2.

3 The singular quartic recovers the axes

Lemma 3.1 (Singular locus of the Hankel determinant)**.** *Over $\mathbf{F}_{11}$, the singular scheme of $H = 0$ is the rational normal quartic R.*

Proof. The five partial derivatives are

$$\begin{aligned} z_2z_4 - z_3^2, &\qquad 2(z_2z_3 - z_1z_4), \\ z_0z_4 + 2z_1z_3 - 3z_2^2, &\qquad 2(z_1z_2 - z_0z_3), \\ z_0z_2 - z_1^2. & \end{aligned}$$

Let I_R be the ideal of the six 2-by-2 minors of

$$\begin{pmatrix} z_0 & z_1 & z_2 & z_3 \\ z_1 & z_2 & z_3 & z_4 \end{pmatrix} .$$

These minors define the rational normal quartic scheme. Four are scalar multiples of the first, second, fourth, and fifth displayed derivatives. Write the other two as

$$A = z_0z_4 - z_1z_3, \qquad B = z_1z_3 - z_2^2.$$

The middle derivative is $A+3B$, so the Jacobian ideal J is contained in I_R. Conversely, on $D(z_2)$, the identity

$$z_2A = z_0(z_2z_4 - z_3^2) + z_3(z_0z_3 - z_1z_2)$$

puts A, and then B, in J_{z_2}. On $D(z_0)$,

$$z_0B = z_1(z_0z_3 - z_1z_2) - z_2(z_0z_2 - z_1^2)$$

does the same; the symmetric identity handles $D(z_4)$. These three opens cover the projective Jacobian scheme, since $z_0 = z_2 = z_4 = 0$ and the outer two derivatives force $z_1 = z_3 = 0$. Hence the saturated Jacobian ideal equals I_R, proving the scheme-theoretic assertion. □

The group G acts on $R \cong \mathbf{P}^1$. For $x \in R(\mathbf{F}_{11})$, its stabilizer lies in the order-55 Borel subgroup of $\mathrm{PSL}_2(11)$, hence has order dividing $\gcd(60, 55) = 5$. A Sylow 5-subgroup is split and fixes two points. Distinct Sylow 5-subgroups cannot share a fixed point, since the point stabilizer has order at most five. The six Sylow subgroups therefore supply $6 \cdot 2 = 12$ distinct points, exhausting $R(\mathbf{F}_{11})$. Every point stabilizer is C_5, the action is transitive, and the six fixed-point pairs form

$$G/N_G(C_5) \cong A_5/D_5.$$

The stabilizer of each pair of the original matching is also D_5, and this subgroup is self-normalizing in G. Hence it fixes a unique matching pair in the transitive G/D_5 action. Sending a singular point to the matching pair fixed by the normalizer of its stabilizer therefore gives the intrinsic map

$$A_5/C_5 \longrightarrow A_5/D_5, \qquad gC_5 \longmapsto gN_G(C_5),$$

with fibres of size two. This proves the axis-compatibility clause of Theorem 1.2.

Lemma 3.2 (The unoriented conference companion)**.** *The recovered G-set $\widehat{X}_M \cong A_5/D_5$ determines a unique G-invariant order-six conference switching class up to opposite, and hence a unique projective triangle-cubic line $[C_M]$.*

Proof. Normalize a symmetric conference signing by switching so that its first row is positive. Orthogonality with that row forces every other row to have two positive and two negative entries away from the first coordinate and its diagonal. The negative entries therefore form a simple two-regular graph on five vertices, hence a pentagon. All pentagons are relabelled copies of one another; complementing the pentagon gives the opposite switching class. The natural six-point A_5-action preserves this class, and its two orientations give opposite triangle cubics on the augmentation. More precisely, the automorphism group of an oriented order-six conference class is the transitive A_5. Identify it with the recovered $G \le S(\widehat{X}_M)$. Two such identifications differ through

$$N_{S_6}(G)/G \cong C_2,$$

and the nontrivial coset exchanges the class with its opposite. Hence the unordered opposite pair, and therefore its projective triangle-cubic line, is intrinsic to the fixed recovered G-set. This classical uniqueness and pentagon representative also appear in Bussemaker–Mathon–Seidel [1, pp. 12, 21, and Table 9 on p. 83]; our switching convention follows Goethals–Seidel [3, §§1–3]. □

Paper I's corollary *Nodes, symmetry, and integral commutant* proves that this triangle cubic has exactly six ordinary nodes in projective-frame position [7]. Since H_M is singular along a curve, the two projective cubic lines are distinct. This completes Theorem 1.2.

4 The marked companion pencil

Use the reconstructed identification of Theorem 1.2 to transport W_5 and $[H_M]$ to the augmentation A_M. The character formula for a symmetric cube gives

$$\dim(\mathrm{Sym}^3(A_M^*)^G) = 2.$$

Thus C_M and H_M, already known to be independent, span the full invariant pencil. Take the odd normalizer permutation (3 4 5 6) on the six axes. In the basis $e_i - e_6$, its augmentation matrix is

$$Q = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ -1 & -1 & -1 & -1 & -1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Substitution of Q in the two complete coefficient vectors gives

$$C_M \longmapsto -C_M, \qquad H_M \longmapsto 8C_M + H_M.$$

Every other odd normalizer element differs from this one by an element of G, which fixes the invariant pencil pointwise. This proves Proposition 1.3. A direct Jacobian calculation gives the visible rational singular members

$$\begin{array}{ccc} [1:0] & 6 & A_5/D_5, \\ [1:3] & 10 & A_5/S_3, \\ [0:1],\ [1:7] & 12 & A_5/C_5. \end{array}$$

The last two entries count the rational points on two singular quartics; they are not twelve-node cubics.

Proof of Theorem 1.4. On the ordered basis (C_M, H_M), the calculation above gives

$$q = \begin{pmatrix} -1 & 8 \\ 0 & 1 \end{pmatrix}.$$

Thus the (-1)-eigenspace is exactly $\mathbf{F}_{11}C_M$, while $\Delta H_M = 8C_M \neq 0$. Since each chordal line is one-dimensional, Δ restricts to an isomorphism from the selected line to the conference line. The other chordal line is qL; applying $q^2 = 1$ on $\mathcal{I}$ shows that Δ is nonzero there as well. The displayed forward and reverse formulas are therefore defined and are inverse. Linearity gives $\Delta(-h) = -\Delta h$, proving compatibility with sheet and conference reversal. Finally, replacing the odd representative q by gq, with $g \in G$, changes nothing on $\mathcal{I}$, because every element of G fixes that space pointwise. □

Remark 4.1 (Unmarked boundary). The oriented round trip is exact at its stated marked level. If the selected chordal line L is forgotten, the conference package retains only the unordered pair $\{L, qL\}$; it cannot choose one member. Thus Papers I and III enter the round trip only relative to a declared companion marking, whereas Paper II supplies that marking through its projected tensor.

5 The three marked transports and their return

We now state exactly what the round trip returns to each earlier paper. A *marked companion package* is

$$\mathcal{P} = (X, \rho : G \hookrightarrow S_X, A_X, L, h, c), \qquad c = 8^{-1}(q-1)h,$$

where S_X is the symmetric group of X, A_X is the augmentation of the marked six-set, $L = \mathbf{F}_{11}h$ is one selected chordal line, and c is the normalized conference generator. We allow exactly the relabellings declared in the source papers; signs and scales not quotiented there remain part of the data. Thus simultaneous negation $(h, c) \mapsto (-h, -c)$ is the orientation involution, not a relabelling. Recording X also transports the normalizer coset defining q. For Paper III, every component of $\mathfrak{m}_{\mathrm{III}}$ remains marked, including the ordered axes, chart lift and its scale, outer/Petersen labels, and cross-identification.

For Paper II, the bridge marking includes the deterministic normalized intertwiner T of Section 2; this is what transports an actual sheet generator, rather than only its projective line, to A. The coefficient normalization 3^{-1} is part of this fixed bridge convention and is transported with it; it is not recomputed after a relabelling. Papers I and III instead begin with a conference signing and must add the choice of L. Their sign matrices have entries $\{\pm 1\}$, so reduction modulo eleven and lifting back to signs are inverse; no integral lattice is being inferred from equality of abstract rational algebras.

Corollary 5.1 (Source-by-source marked return)**.** *Relative to the markings in the following table, each of Papers I–III maps to the corresponding decorated marked companion package and is recovered from it at exactly the stated output level.*

paper	*retained source output*	*target decoration*	*inverse returns*
I	*conference switching class, axis carrier, orientation, and selected chordal line L*	*the marked core $\mathcal{P}$*	*the same class, carrier, orientation, and L, modulo switching and relabelling*
II	*five-isotypic projected tensor with chosen sheet sign, marked axes, and T*	*W_5, T, and the fixed scalar convention*	*the same projected tensor, sheet sign, six-axis carrier, and T*
III	*relative conference source, selected L, and full bridge datum*	$\mathfrak{m}_{\mathrm{III}}$	*the same relative conference source, orientation, L, and bridge datum*

The table records the extra marking carried through each transport and the output recovered by its inverse. Every forward and inverse construction is equivariant under the allowed relabellings. Hence, for every source package arising from Papers I–III with the stated marking, forward followed by inverse recovers the source output, and inverse followed by forward recovers the marked companion package. No inverse is claimed outside this image.

Proof. Paper I's *Support cubic and golden continuation* and its stable six-node corollary recover the conference signing, orientation, and axis carrier [7]. Paper III's *Relative marked orientation bridge* supplies the same conference source relative to its explicitly declared bridge datum [8]. Given the added line L, Theorem 1.4 recovers the unique $h \in L$ satisfying $(q-1)h = 8c$. Returning forgets that derived generator and leaves every source output unchanged; the base marking was never forgotten.

For Paper II, *Balanced sheets and cubic-first orientation* supplies the signed generator, and Theorem 1.2 recovers the axis carrier. Theorem 1.4 then gives c. The reverse formula recovers the same h. In the notation of Section 2, the exact forward and reverse tensor maps are

$$h = 3^{-1}\,\mathrm{Pol}\Big(\mathrm{Sym}^3(BT^{-1})\nu\Big)\,, \qquad \nu = \mathrm{Sym}^3(TB^{-1})\,\mathrm{Pol}^{-1}(3h).$$

Thus the pivot normalization, polynomialization, self-duality, and intertwiner are all undone. Axis relabellings induce every map in these formulas, so the formulas are equivariant. The asserted returns are therefore the displayed inverse formulas together with the stable recovery statements just cited. □

The corollary deliberately returns neither the full Paper-II quotient nor the global Paper-III cover. Paper III's chart lift is carried unchanged inside its fixed bridge datum; it is not reconstructed from the companion core. Paper IV remains the independent minimum-word reconstruction branch and does not enter these transports.

Corollary 5.2 (Stabilizer-stratified base change)**.** *Let $K/\mathbf{F}_{11}$ be any field extension, and base-change the fixed marked Paper-II package to K. Put $R_K = \mathrm{Sing}(H_{M,K})$, and for $P \leq G$ write R_K^P for its fixed-point subscheme. Then*

$$Z_{5,K} = \coprod_{P\in\mathrm{Syl}_5(G)} R_K^P$$

is split finite étale of degree twelve. Its geometric points have exact stabilizer C_5, and the normalizer map is a canonical G-equivariant double cover

$$Z_{5,K} \longrightarrow \mathrm{Syl}_5(G)_K \cong (A_5/D_5)_K.$$

Here $\mathrm{Syl}_5(G)_K$ is the constant K-scheme on the six Sylow 5-subgroups. Its six fibres recover the base-changed axis carrier. The invariant pencil, the marked $q-1$ correspondence, and the Paper-II tensor forward and reverse maps commute with this base change and with every further extension of K.

Proof. Since $11 \nmid 60$, the Reynolds idempotent splits invariants, so

$$(\mathrm{Sym}^3(A_M^*)^G) \otimes_{\mathbf{F}_{11}} K \cong \mathrm{Sym}^3(A_{M,K}^*)^G.$$

The invariant pencil therefore remains two-dimensional. The coefficient identity with the Hankel determinant and the equality of its saturated Jacobian scheme with the rational normal quartic are identities over $\mathbf{F}_{11}$; flat base change gives R_K scheme-theoretically.

The action of each Sylow 5-subgroup on R is split over $\mathbf{F}_{11}$ and tame. A generator has two distinct eigenlines on $R \cong \mathbf{P}^1$, so its fixed-point subscheme is reduced, split, and has degree two. The stabilizer argument proving the axis-compatibility clause of Theorem 1.2 shows that the six fixed pairs are disjoint and that every one of their points has exact stabilizer that Sylow subgroup. Thus $Z_{5,K}$ is the scalar extension of the constant $\mathbf{F}_{11}$-scheme $(G/C_5)_{\mathbf{F}_{11}}$. The map $(G/C_5)_K \to (G/N_G(C_5))_K = (G/D_5)_K$ is finite of degree two and also commutes with base change. Finally, the pencil matrix, the two tensor formulas, and their composites are matrix identities over $\mathbf{F}_{11}$; the scalars 3 and 8 remain units in K. Hence the marked $q-1$ inverse and the Paper-II tensor inverse remain exact. □

This does not create a new matching problem on $\mathbf{P}^1(\mathbf{F}_{11^m})$. Indeed, the whole singular quartic has 11^m+1 rational points over $\mathbf{F}_{11^m}$; the intrinsic selector is the fixed degree-twelve scheme $Z_{5,\mathbf{F}_{11^m}}$, not the full rational locus.

6 Verification and trust boundary

The transport matrices and coefficient identities are finite computer-assisted leaves of the proof. The paper-owned standard-library checker

`verification/evidence/paper_ii_chordal_axis.py`

has schema `paper-v-chordal-axis-v2`. It reconstructs the matching quotient from Paper II's frozen arithmetic, computes the central projector and both intertwiner spaces, checks all cubic coefficients, enumerates the rational singular loci, and determines the pencil action. Its exact output is the adjacent JSON file. From the repository root, the replay is

```
python3 papers/clebsch-round-trip/verification/evidence/
        paper_ii_chordal_axis.py -check
```

and returns `CHECK OK (NO_MATCH)`; `NO_MATCH` is the proved failure of the literal conference-line identification, not a failed check. The JSON names, sizes, and hashes both frozen inputs. Their SHA-256 values begin `d2100485` for the matching-module source and `d45512c9` for the orbit-scout data; the unabbreviated values are part of the machine-readable proof interface. A separately written cold replay reproduced the normalized coefficient vector; its full byte SHA-256 is recorded in the adjacent JSON certificate. Neither execution substitutes for Lemma 3.1 or the stabilizer argument proving the axis-compatibility clause of Theorem 1.2.

7 Conclusion

Paper II's cubic shadow reaches the golden six-axis carrier without being the conference cubic. Projection gives a chordal cubic; its singular quartic gives twelve points; exact stabilizers pair them into the original six axes. The conference and chordal members thus retain one carrier through different singular shadows. On their invariant pencil, $q - 1$ carries the selected sheet generator to the conference orientation and is invertible on either selected chordal line. The oriented return is exact, but the companion choice is irreducible: an unmarked conference package remembers the pair of chordal lines without selecting one.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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